

Week 5

Q1. Write a C++ program to enter elements in the array and display the array elements.

```
#include <iostream>
using namespace std;

int main() {
    int n, arr[100];
    cout << "Enter the no. of elements: ";
    cin >> n;
    cout << "Insert the elements: " << endl;
    for (int i=0; i<n; i++) {
        cin >> arr[i];
    }
    cout << "Elements you inserted in the array: ";
    for (int i=0; i<n; i++) {
        cout << arr[i] << " ";
    }
    return 0;
}
```

```
Enter the no. of elements: 5
Insert the elements:
10
2
30
4
50
Elements you inserted in the array: 10 2 30 4 50
-----
Process exited after 15.71 seconds with return value 0
Press any key to continue . . . |
```

Q2. Write a C++ program to find the sum of all the array elements.

```
#include <iostream>

using namespace std;
int main() {
    int sum = 0;

    int arr[] = {1, 2, 3, 4, 5};
    int size = sizeof(arr)/sizeof(arr[0]);
    for(int i=0; i<size; i++) {
        sum += arr[i];
    }
    cout << "Sum of all elements = " << sum;
    return 0;
}
```

```
Sum of all elements = 15
-----
Process exited after 1.144 seconds with return value 0
Press any key to continue . . . |
```

Q3. Write a C++ program to find the length of the array.

```
#include <iostream>

using namespace std;
int main() {
    int arr[] = {1, 2, 3, 4, 5};
    int size = sizeof(arr)/sizeof(arr[0]);
    cout << "Size of given Array = " << size;
    return 0;
}
```

```
Size of given Array = 5
-----
Process exited after 1.152 seconds with return value 0
Press any key to continue . . . |
```

Q4. Write a C++ program to find the second-largest integer in a list of integers.

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter number of elements: ";
    cin >> n;

    int largest = INT_MIN;
    int second = INT_MIN;

    cout << "Enter the integers:\n";

    for (int i = 0; i < n; i++) {
        int x;
        cin >> x;

        if (x > largest) {
            second = largest;
            largest = x;
        } else if (x > second && x != largest) {
            second = x;
        }
    }

    if (second == INT_MIN) {
        cout << "There is no second-largest distinct integer." << endl;
    } else {
        cout << "Second-largest integer: " << second << endl;
    }

    return 0;
}
```

```
Enter number of elements: 5
Enter the integers:
10
100
60
2
519
Second-largest integer: 100

-----
Process exited after 21.45 seconds with return value 0
Press any key to continue . . . |
```

Q5. Write a C++ Program to reverse the position of the array element (Hint: First element to the last element.)

```
#include <iostream>
using namespace std;

int main() {
    int arr[] = {1, 3, 5, 7, 9};
    int size = sizeof(arr)/sizeof(arr[0]);

    for(int i=0; i<size/2; i++) {
        int temp = arr[i];
        arr[i] = arr[size-i-1];
        arr[size-i-1] = temp;
    }

    for (int i=0; i<size; i++) {
        cout << arr[i] << " ";
    }

    return 0;
}
```

```
9 7 5 3 1
-----
Process exited after 1.659 seconds with return value 0
Press any key to continue . . . |
```

Q6. Write a C++ program to perform the following: a. Addition of two matrices b. Multiplication of two matrices

```
#include <iostream>
using namespace std;

int main() {
    int r1, c1, r2, c2;

    cout << "Enter rows and columns of first matrix: ";
    cin >> r1 >> c1;

    cout << "Enter rows and columns of second matrix: ";
    cin >> r2 >> c2;

    int A[r1][c1], B[r2][c2], sum[r1][c1], product[r1][c2];

    cout << "Enter elements of first matrix:\n";
    for (int i = 0; i < r1; i++) {
        for (int j = 0; j < c1; j++) {
            cin >> A[i][j];
        }
    }

    cout << "Enter elements of second matrix:\n";
    for (int i = 0; i < r2; i++) {
        for (int j = 0; j < c2; j++) {
            cin >> B[i][j];
        }
    }

    if (r1 == r2 && c1 == c2) {
        for (int i = 0; i < r1; i++) {
            for (int j = 0; j < c1; j++) {
                sum[i][j] = A[i][j] + B[i][j];
            }
        }

        cout << "\nAddition of two matrices:\n";
        for (int i = 0; i < r1; i++) {
            for (int j = 0; j < c1; j++) {
                cout << sum[i][j] << " ";
            }
            cout << endl;
        }
    } else {
        cout << "\nMatrix addition is not possible.\n";
        cout << "Both matrices must have the same dimensions.\n";
    }
}
```

```
    if (c1 == r2) {
        for (int i = 0; i < r1; i++) {
            for (int j = 0; j < c2; j++) {
                product[i][j] = 0;
            }
        }

        for (int i = 0; i < r1; i++) {
            for (int j = 0; j < c2; j++) {
                for (int k = 0; k < c1; k++) {
                    product[i][j] += A[i][k] * B[k][j];
                }
            }
        }

        cout << "\nMultiplication of two matrices:\n";
        for (int i = 0; i < r1; i++) {
            for (int j = 0; j < c2; j++) {
                cout << product[i][j] << " ";
            }
            cout << endl;
        }
    } else {
        cout << "\nMatrix multiplication is not possible.\n";
        cout << "Columns of first matrix must equal rows of second matrix.\n";
    }

    return 0;
}
```

```

Enter rows and columns of first matrix: 2 2
Enter rows and columns of second matrix: 2 2
Enter elements of first matrix:
10
20
30
40
Enter elements of second matrix:
1
1
1
1

Addition of two matrices:
11 21
31 41

Multiplication of two matrices:
30 30
70 70

-----
Process exited after 32.15 seconds with return value 0
Press any key to continue . . . |

```

Q7. Write a C++ program to count and display positive, negative, odd, and even numbers in an array.

```

#include <iostream>
using namespace std;

int main() {
    int n, positive = 0, negative = 0, odd = 0, even = 0;

    cout << "Enter the number of elements: ";
    cin >> n;

    int arr[n];

    cout << "Enter " << n << " elements:\n";
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }

    cout << "\nPositive numbers: ";
    for (int i = 0; i < n; i++) {
        if (arr[i] > 0) {
            cout << arr[i] << " ";
            positive++;
        }
    }
    cout << "\nCount = " << positive;

    cout << "\n\nNegative numbers: ";
    for (int i = 0; i < n; i++) {
        if (arr[i] < 0) {
            cout << arr[i] << " ";
            negative++;
        }
    }
    cout << "\nCount = " << negative;
}

```

```

    cout << "\n\nOdd numbers: ";
    for (int i = 0; i < n; i++) {
        if (arr[i] % 2 != 0 && arr[i] > 0) {
            cout << arr[i] << " ";
            odd++;
        }
    }
    cout << "\nCount = " << odd;

    cout << "\n\nEven numbers: ";
    for (int i = 0; i < n; i++) {
        if (arr[i] % 2 == 0 && arr[i] > 0) {
            cout << arr[i] << " ";
            even++;
        }
    }
    cout << "\nCount = " << even;

    return 0;
}

```

```
Enter the number of elements: 5
Enter 5 elements:
-1
2
0
-3
5

Positive numbers: 2 5
Count = 2

Negative numbers: -1 -3
Count = 2

Odd numbers: 5
Count = 1

Even numbers: 2
Count = 1
-----
Process exited after 15.58 seconds with return value 0
Press any key to continue . . . |
```

Q8. Write a C++ program to merge two sorted arrays into another array in sorted order.

```

#include <iostream>
using namespace std;

int main() {
    int n, m;

    cout << "Enter size of first array: ";
    cin >> n;

    int A[100];
    cout << "Enter elements of first sorted array: ";
    for (int i = 0; i < n; i++) {
        cin >> A[i];
    }

    cout << "Enter size of second array: ";
    cin >> m;

    int B[100];
    cout << "Enter elements of second sorted array: ";
    for (int i = 0; i < m; i++) {
        cin >> B[i];
    }

    int C[200];
    int i = 0, j = 0, k = 0;

    while (i < n && j < m) {
        if (A[i] <= B[j]) {
            C[k] = A[i];
            i++;
        } else {
            C[k] = B[j];
            j++;
        }
        k++;
    }
}

```

```

    while (i < n) {
        C[k] = A[i];
        i++;
        k++;
    }

    while (j < m) {
        C[k] = B[j];
        j++;
        k++;
    }

    cout << "Merged sorted array: ";
    for (int i = 0; i < n + m; i++) {
        cout << C[i] << " ";
    }

    return 0;
}

```

```

Enter size of first array: 5
Enter elements of first sorted array: 1 3 5 7 9
Enter size of second array: 3
Enter elements of second sorted array: 2 4 6
Merged sorted array: 1 2 3 4 5 6 7 9
-----
Process exited after 17.88 seconds with return value 0
Press any key to continue . . . |

```

Q9. Write a C++ program to find the frequency of a particular number in a list of integers.

```
#include <iostream>
using namespace std;

int main() {
    int n, target, frequency = 0;
    cout << "Enter the number of elements: ";
    cin >> n;
    int num[n];
    cout << "Enter the integers: ";
    for (int i = 0; i < n; i++) {
        cin >> num[i];
    }

    cout << "Enter the number to find: ";
    cin >> target;

    cout << "Enter the integers: ";
    for (int i = 0; i < n; i++) {
        if (num[i] == target) {
            frequency++;
        }
    }

    cout << "Frequency of " << target << " = " << frequency << endl;

    return 0;
}
```

```
Enter the number of elements: 5
Enter the integers: 1 1 2 1 3
Enter the number to find: 1
Enter the integers: Frequency of 1 = 3
```

```
-----
Process exited after 12.42 seconds with return value 0
Press any key to continue . . . |
```